

Модуль 9: Введение в Ansible (Управление конфигурациями)

- 1) Задача (Установка): Установить Ansible. (e.g., `pip install ansible`).

```
root@Cetus:/home/ikomarov# ansible --version
ansible [core 2.18.10]
  config file = /etc/ansible/ansible.cfg
  configured module search path = ['/root/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  ansible collection location = /root/.ansible/collections:/usr/share/ansible/collections
  executable location = /usr/bin/ansible
  python version = 3.12.3 (main, Jan  8 2026, 11:30:50) [GCC 13.3.0] (/usr/bin/python3)
  jinja version = 3.1.2
  libyaml = True
```

- 2) Задача (Ad-Нос - ping): Проверить связь: `ansible local -i inventory.ini -m ping`.

```
root@Cetus:/opt/basic/module9# root@Cetus:/opt/basic/module9# ansible local -i inventory.ini -m ping
[WARNING]: Platform linux on host localhost is using the discovered Python interpreter at /usr/bin/python
information.
localhost | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.12"
  },
  "changed": false,
  "ping": "pong"
}
```

- 3) Задача (Ad-Нос - setup): Собрать "факты" о машине: `ansible local -i inventory.ini -m setup`.

```
root@Cetus:/opt/basic/module9# ansible local -i inventory.ini -m setup
[WARNING]: Platform linux on host localhost is using the discovered Python interpreter at /usr/bin/pyt
information.
localhost | SUCCESS => {
  "ansible_facts": {
    "ansible_all_ipv4_addresses": [
      "172.20.0.1",
      "172.19.0.1",
      "172.18.0.1",
      "10.42.0.0",
      "10.0.2.15",
      "10.42.0.1",
      "172.21.0.1",
      "172.17.0.1"
    ],
    "ansible_all_ipv6_addresses": [
      "fe80::5878:e0ff:fed6:a45d",
      "fe80::8475:abff:fea8:d4df",
      "fe80::c4c0:93ff:fef1:b6b0",
      "fe80::f897:89ff:fe72:af56",
      "fe80::30d8:9ff:fe6f:66cb",
      "fd17:625c:f037:2:a00:27ff:fe79:1556",
      "fe80::a00:27ff:fe79:1556",
      "fe80::4cb:59ff:fe67:9d02",
      "fe80::7801:feff:fed2:2ee4"
    ],
    "ansible_apparmor": {
      "status": "enabled"
    },
    "ansible_architecture": "x86_64",
```

- 4) Задача (Ad-Хос - apt): Установить Nginx: ansible local -i inventory.ini -m apt -a "name=nginx state=present" --become.

```
root@Cetus:/opt/basic/module9# ansible local -i inventory.ini -m apt -a "name=nginx state=present" --become
[WARNING]: Platform linux on host localhost is using the discovered Python interpreter at /usr/bin/python3.12, but future installation of
information.
localhost | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.12"
  },
  "cache_update_time": 1770230262,
  "cache_updated": false,
  "changed": true,
  "stderr": "\nRunning kernel seems to be up-to-date.\n\nNo services need to be restarted.\n\nNo containers need to be restarted.\n\nNo
  "stderr_lines": [
    "",
    "Running kernel seems to be up-to-date.",
    "",
    "No services need to be restarted.",
    "",
    "No containers need to be restarted.",
    "",
    "No user sessions are running outdated binaries.",
    "",
    "No VM guests are running outdated hypervisor (qemu) binaries on this host."
  ],
  "stdout": "Reading package lists...\nBuilding dependency tree...\nReading state information...\nThe following additional packages will
led, 0 to remove and 126 not upgraded.\nNeed to get 564 kB of archives.\nAfter this operation, 1596 kB of additional disk space will be use
amd64 nginx amd64 1.24.0-2ubuntu7.5 [520 kB]\nPreconfiguring packages ...\nFetched 564 kB in 0s (3131 kB/s)\nSelecting previously unselecte
. 25%\r(Reading database ... 30%\r(Reading database ... 35%\r(Reading database ... 40%\r(Reading database ... 45%\r(Reading database ... 50
base ... 85%\r(Reading database ... 90%\r(Reading database ... 95%\r(Reading database ... 100%\r(Reading database ... 155323 files and dire
lected package nginx.\r\nPreparing to unpack .../nginx_1.24.0-2ubuntu7.5_amd64.deb ... \r\nUnpacking nginx (1.24.0-2ubuntu7.5) ... \r\nSettin
inx (1.24.0-2ubuntu7.5) ... \r\n * Upgrading binary nginx      \u001b[80G \r\u001b[74G[ OK ]\r\nProcessing triggers for man-db (2.12.0-4bu
  "stdout_lines": [
    "Reading package lists...",
    "Building dependency tree...",
    "Reading state information...",
    "The following additional packages will be installed:",
    "  nginx-common",
    "Suggested packages:",
    "  fcgiwrap nginx-doc ssl-cert",
    "The following NEW packages will be installed:",
    "  nginx nginx-common",
    "0 upgraded, 2 newly installed, 0 to remove and 126 not upgraded.",
    "Need to get 564 kB of archives.",
    "After this operation, 1596 kB of additional disk space will be used.",
    "Get:1 http://ru.archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx-common all 1.24.0-2ubuntu7.5 [43.4 kB]",
    "Get:2 http://ru.archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx amd64 1.24.0-2ubuntu7.5 [520 kB]",
    "Preconfiguring packages ...",
    "Fetched 564 kB in 0s (3131 kB/s)",
    "Selecting previously unselected package nginx-common.",
    "(Reading database ... ",
    "(Reading database ... 5%",
  ],
}
```

- 5) Задача (Идемпотентность): Запустить команду из Задачи 6 еще раз.

```
root@Cetus:/opt/basic/module9# nginx -v
nginx version: nginx/1.24.0 (Ubuntu)
root@Cetus:/opt/basic/module9# ansible local -i inventory.ini -m apt -a "name=nginx state=present" --become
[WARNING]: Platform linux on host localhost is using the discovered Python interpreter at /usr/bin/python3.12, but future installation of
information.
localhost | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.12"
  },
  "cache_update_time": 1770230262,
  "cache_updated": false,
  "changed": false
}
root@Cetus:/opt/basic/module9#
```

- 6) Задача (Ad-Нос - file): Создать папку: `ansible local -i inventory.ini -m file -a "path=/tmp/test-ansible state=directory"`.

Задача (Ad-Нос - copy): Скопировать файл: `ansible local -i inventory.ini -m copy -a "src=inventory.ini dest=/tmp/test-ansible/inventory.bak"`.

```
root@Cetus:/opt/basic/module9# ansible local -i inventory.ini -m file -a "path=/tmp/test-ansible state=directory"
[WARNING]: Platform linux on host localhost is using the discovered Python interpreter at /usr/bin/python3.12, but future installation of another Python interpreter could chan
Information.
localhost | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.12"
  },
  "changed": true,
  "gid": 0,
  "group": "root",
  "mode": "0755",
  "owner": "root",
  "path": "/tmp/test-ansible",
  "size": 4096,
  "state": "directory",
  "uid": 0
}
root@Cetus:/opt/basic/module9# ls /tmp
snap-private-tmp                                systemd-private-c882a6a2af694f47957580480be4675b-polkit.service-0L2UrQ      systemd-private-
systemd-private-c882a6a2af694f47957580480be4675b-fwupd.service-mRz1sp          systemd-private-c882a6a2af694f47957580480be4675b-systemd-logind.service-cHCpF8  systemd-private
systemd-private-c882a6a2af694f47957580480be4675b-ModemManager.service-Exe34x  systemd-private-c882a6a2af694f47957580480be4675b-systemd-resolved.service-LCZMGh test-ansible
root@Cetus:/opt/basic/module9# ansible local -i inventory.ini -m copy -a "src=inventory.ini dest=/tmp/test-ansible/inventory.bak"
[WARNING]: Platform linux on host localhost is using the discovered Python interpreter at /usr/bin/python3.12, but future installation of another Python interpreter could chan
Information.
localhost | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.12"
  },
  "changed": true,
  "checksum": "381cbc692c31932bbd37434e8643fff280ef193d",
  "dest": "/tmp/test-ansible/inventory.bak",
  "gid": 0,
  "group": "root",
  "md5sum": "3ee28d24745116de87806c0e46a87ea0",
  "mode": "0644",
  "owner": "root",
  "size": 44,
  "src": "/root/.ansible/tmp/ansible-tmp-1770231262.7338839-5818-16006923386405/.source.bak",
  "state": "file",
  "uid": 0
}
root@Cetus:/opt/basic/module9#
```